$c_p := 1006 \ \frac{Дж}{кг \cdot K}$ $\eta_t := 0.75$ СП 131.13330.2025 Строительна климатология при φ=55%

$\eta_к := 0.75$

$R_a := 287.05 \ \frac{Дж}{кг \cdot K}$ $\eta_м := 0.98$ $d_{H0} := 12.2 \ \frac{г}{кг}$

$k := 1.4$ $p_{H0} := 101325 \ Па$ $\rho_{H0} := 1.3 \ \frac{кг}{м^3}$

$M_1 := 0.8$ $H_1 := 200 \ м$ $T_{ocH01} := 313.1 \cdot K$ $Q_{co1} := 3800 \ Вт$

$M_2 := 1.8$ $H_2 := 15000 \ м$ $T_{ocH02} := 313.1 \cdot K$ $Q_{co2} := 4400 \ Вт$

$M_3 := 2.2$ $H_3 := 12000 \ м$ $T_{ocH03} := 313.1 \cdot K$ $Q_{co3} := 4800 \ Вт$

Параметры окружающей среды для режимов: $T_{oc2} := 216.65 \ K$ - Стратосфера

$T_{oc1} := T_{ocH01} - 0.0065 \cdot H_1 = 311.8 \ K$

$T_{oc3} := T_{ocH03} - 0.0065 \cdot H_3 = 235.$ K

$p_{oc1} := p_{H0} \cdot \left(1 - \frac{H_1}{44300}\right)^{5.256} = 9.894 \times 10^4 \ Па$ $p_{oc2} := 22690 \cdot e^{-\left(\frac{H_2 - 11000}{6340}\right)} = 1.207 \times 10^4 \ Па$

$p_{oc3} := p_{H0} \cdot \left(1 - \frac{H_3}{44300}\right)^{5.256} = 1.926 \times 10^4 \ Па$ $\rho_{oc1} := \rho_{H0} \cdot \left(1 - \frac{H_1}{44300}\right)^{4.256} = 1.275 \ \frac{г}{м^3}$

$\rho_{oc2} := 0.3649 \cdot e^{-\left(\frac{H_2 - 11000}{6340}\right)} = 0.194 \ \frac{кг}{м^3}$ $\rho_{oc3} := \rho_{H0} \cdot \left(1 - \frac{H_3}{44300}\right)^{4.256} = 0.339 \ \frac{г}{м^3}$

$d_{oc1} := d_{H0} \cdot 10^{-\left(\frac{H_1}{5000}\right)} = 11.127 \ \frac{г}{кг}$

Параметры торможения:

$T_{tor1} := T_{oc1} \cdot \left(1 + 0.2 \cdot M_1^2\right) = 351.76 \cdot K$ $p_{tor1} := p_{oc1} \cdot \left(1 + 0.2 \cdot M_1^2\right)^{3.5} = 1.508 \times 10^5 \ Па$

Примем: $\omega := 15 \cdot deg$ $\alpha := 50 \cdot deg$

Given

$$\tan(\omega) = \frac{M_2^{\,2}\cdot(\sin(\alpha))^2 - 1}{\left[1 + \left(\frac{k-1}{2}\right)\cdot M_2^{\,2}\right]\cdot(\tan(\alpha)) + M_2^{\,2}\cdot(\sin(\alpha))\cdot(\cos(\alpha))}$$

$\text{Find}(\alpha) = 0.896$

$\alpha_2 := 51.34\cdot\text{deg} \qquad \alpha_3 := 41.27\cdot\text{deg} \qquad \omega := 15\cdot\text{deg}$

$$p_{occ2} := \left[\left(\frac{2\cdot k}{k+1}\right)\cdot M_2^{\,2}\cdot\sin(\alpha_2)^2 - \frac{k-1}{k+1}\right]\cdot p_{oc2} = 2.582\times 10^4 \ \text{Па}$$

$$T_{occ2} := \left[\left(\frac{2}{k+1}\right)\cdot\left[\frac{1}{M_2^{\,2}\cdot(\sin(\alpha_2))^2}\right] + \frac{k-1}{k+1}\right]\cdot\left[\left(\frac{2k}{k+1}\right)\cdot M_2^{\,2}\cdot(\sin(\alpha_2))^2 - \frac{(k-1)}{(k+1)}\right]\cdot T_{oc2} = 272.608\ \text{K}$$

$$M_{sk2} := \sqrt{\frac{\left(\frac{2}{k-1}\right) + M_2^{\,2}\cdot(\sin(\alpha_2))^2}{\left(\frac{2\cdot k}{k-1}\right)\cdot\left[M_2^{\,2}\cdot(\sin(\alpha_2))^2 - 1\right]\cdot\sin(\alpha_2 - \omega)}} = 1.313$$

$$T_{tor2} := T_{occ2}\cdot\left[1 + \frac{(k-1)\cdot M_{sk2}^{\,2}}{2}\right] = 366.588\ \text{K}$$

$$p_{tor2} := p_{occ2}\cdot\left(1 + \frac{k-1}{2}\cdot M_{sk2}^{\,2}\right)^{\frac{k}{k-1}} = 7.28\times 10^4 \ \text{Па}$$

$$p_{occ3} := \left[\left(\frac{2\cdot k}{k+1}\right)\cdot M_3^{\,2}\cdot\sin(\alpha_3)^2 - \frac{k-1}{k+1}\right]\cdot p_{oc3} = 4.41\times 10^4 \ \text{Па}$$

$$T_{occ3} := \left[\left(\frac{2}{k+1}\right)\cdot\left[\frac{1}{M_3^{\,2}\cdot(\sin(\alpha_3))^2}\right] + \frac{k-1}{k+1}\right]\cdot\left[\left(\frac{2k}{k+1}\right)\cdot M_3^{\,2}\cdot(\sin(\alpha_3))^2 - \frac{(k-1)}{(k+1)}\right]\cdot T_{oc3} = 302.862\ \text{K}$$

$$M_{sk3} := \sqrt{\frac{\left(\frac{2}{k-1}\right) + M_3^{\,2}\cdot(\sin(\alpha_3))^2}{\left(\frac{2\cdot k}{k-1}\right)\cdot\left[M_3^{\,2}\cdot(\sin(\alpha_3))^2 - 1\right]\cdot\sin(\alpha_3 - \omega)}} = 1.44$$

$$T_{tor3} := T_{occ3}\cdot\left[1 + \frac{(k-1)\cdot M_{sk3}^{\,2}}{2}\right] = 428.493\ \text{K}$$

$$p_{tor3} := p_{occ3}\cdot\left(1 + \frac{k-1}{2}\cdot M_{sk3}^{2}\right)^{\frac{k}{k-1}} = 1.486\times 10^{5}\ \text{Па}$$

Расчёт режима 1:

Баланс АТ1:

$$Q_{AT1} = G_1\cdot c_p\cdot\left(T_1 - T_2\right)$$

$$Q_{AT1} = G_{12}\cdot c_p\left(T_5 - T_{12}\right)$$

$$\Delta p_{AT1} := 0.05\cdot p_{tor1} = 7.541\times 10^{3}\ \text{Па}$$

$G_{12} = 0.25G_1$ $\quad \beta := 0.25$ $\quad \varepsilon_1 := 0.85$ - Примем

$$T_2 = T_1 - \varepsilon_1\cdot\beta\cdot\left(T_1 - T_{12}\right)$$

$$T_5 = T_{12} + (1-\beta)\cdot\left(T_1 - T_{12}\right)$$

$T_{12} = T_3$ $\qquad p_2 := p_{tor1} - \Delta p_{AT1} = 1.433\times 10^{5}$ Па

$$T_3 = T_2\cdot\left[1 - \eta_t\left[1 - \left(\frac{p_3}{p_2}\right)^{\frac{(k-1)}{k}}\right]\right]$$

Тогда:

$$T_1 := T_{tor1}$$

$$T_2 = \frac{\left(1-\varepsilon_1\cdot\beta\right)T_1}{1 - \varepsilon_1\cdot\beta\cdot\left[1 - \eta_t\left[1 - \left(\frac{p_3}{p_2}\right)^{\frac{(k-1)}{k}}\right]\right]}$$

$$T_6 = (1-\beta)\cdot T_4 + \beta T_5$$

$$T_7 = T_6\left[1 + \frac{\left(\frac{p_7}{p_6}\right)^{\frac{(k-1)}{k}} - 1}{\eta_к}\right]$$

Баланс АТ2:

$G_4 = (1 - \beta)\cdot G_1$

$T_8 := 323.15\,K \qquad T_{10} := 313.15\ K$ ГОСТ 58247-2018

$$G_{bo} := \frac{Q_{co1}}{c_p\cdot(T_8 - T_{10})} = 0.378 \quad \frac{кг}{с}$$

$G_8 = G_9 = G_{10} = G_{bo}$

Для вентиляора примем:

$$T_9 := T_8\cdot\left[1 + \frac{\pi_{v1}^{\frac{(k-1)}{k}} - 1}{\eta_{v1}}\right] = 330.71\ K \qquad N_{v1} := G_{bo}\cdot c_p\cdot(T_9 - T_8) = 2.873\times 10^3\ Вт$$

$$Q_{AT2} := Q_{co1} + N_{v1} = 6.673\times 10^3\ Вт$$

$$T_4 = T_3 + \frac{Q_{AT2}}{(1 - \beta)\cdot G_1\cdot c_p}$$

$\varepsilon_2 := \varepsilon_1 = 0.85$

Для 6 точки смешения имеем:

$$T_4 = T_3 + \frac{Q_{co1}}{(1 - \beta)\cdot G_1\cdot c_p}$$

$$T_6 = T_3 + \beta\cdot\varepsilon_1\cdot(T_1 - T_3) + \frac{Q_{AT2}}{G_1\cdot c_p}$$

Для вала:

$\eta_м\cdot G_1\cdot c_p\cdot(T_2 - T_3) = G_1\cdot c_p(T_7 - T_6)$

$T_2 - T_3 = (1 - \beta\cdot\varepsilon_1)\cdot(T_1 - T_3)$

Для компрессора:

$$T_7 - T_6 = \left(\frac{T_6}{\eta_к}\right)\cdot\left[\left(\frac{p_7}{p_6}\right)^{\frac{(k-1)}{k}} - 1\right]$$

Примем потери на холодном потоке:

$p_4 = p_5 = 0.9p_3$

Опишем функции

$$a := \frac{k - 1}{k}$$

$$A(p) := 1 - \eta_t \cdot \left[1 - \left(\frac{p}{p_2}\right)^a\right]$$

$$T_2(p) := \frac{(1 - \beta \cdot \varepsilon_1) T_1}{1 - \beta \cdot \varepsilon_1 \cdot A(p)}$$

$$T_3(p) := A(p) \cdot T_2(p)$$

$$T_4(p, G) := T_3(p) + \frac{Q_{AT2}}{(1 - \beta) \cdot G \cdot c_p}$$

$$T_5(p) := T_3(p) + \varepsilon_1 \cdot (T_1 - T_3(p))$$

$$T_6(p, G) := (1 - \beta) \cdot T_4(p, G) + \beta \cdot T_5(p)$$

Решим систему для p3 и G1 относительно AT2 и вала:

$p_3 := 84000$ Па $\quad G_1 := 0.41 \frac{кг}{с}$ - первое прибижение

Given

$0 < p_3 < p_2 \quad G_1 > 0$

$$Q_{AT2} = \varepsilon_2 \cdot c_p \cdot \min\left[G_{bo}, (1 - \beta) \cdot G_1\right] \cdot \left(T_9 - T_3\left(p_3\right)\right)$$

$$\eta_м \cdot \left(T_2\left(p_3\right) - T_3\left(p_3\right)\right) = \frac{T_6\left(p_3, G_1\right)}{\eta_к} \cdot \left[\left(\frac{p_{oc1}}{0.9p_3}\right)^a - 1\right]$$

$$\text{Find}\left(p_3, G_1\right) = \begin{pmatrix} 8.372 \times 10^4 \\ 0.411 \end{pmatrix} \quad p_3 := 8.372 \times 10^4 \quad G_1 := 0.41 \quad \Delta p_{AT2} := 0.1p_3$$

Найдём температуры и давления в точках:

$T_1 = 351.767\ K$ $\quad p_1 := p_{tor1} = 1.508 \times 10^5\ Па$

$T_2(p_3) = 341.918\ K$ $\quad p_2 = 1.433 \times 10^5\ Па$

$T_3(p_3) = 305.422\ K$ $\quad p_3 = 8.372 \times 10^4\ Па$

$T_4(p_3, G_1) = 326.993\ K$ $\quad p_4 := p_3 - \Delta p_{AT2} = 7.53\ Па\ 10^4$

$T_5(p_3) = 344.815\ K$ $\quad p_5 := p_3 - \Delta p_{AT2} = 7.535 \times 10^4\ Па$

$T_6(p_3, G_1) = 331.449\ K$ $\quad p_6 := p_3 - \Delta p_{AT2} = 7.535 \times 10^4\ Па$

$$T_7 := T_6(p_3, G_1) + \left(\frac{T_6(p_3, G_1)}{\eta_к}\right)\cdot\left[\left(\frac{p_{oc1}}{0.9 p_3}\right)^a - 1\right] = 367.22\ K \quad p_7 := p_{oc1} = 9.894 \times 10^4\ Па$$

$T_8 = 323.15\ K$ $\quad p_8 := p_{H0} = 1.013 \times 10^5\ Па$

$T_9 = 330.71\ K$ $\quad p_9 := 1.05 \cdot p_8 = 1.064 \times 10^5\ Па$

$T_{10} = 313.15\ K$ $\quad p_{10} := p_{H0} = 1.013 \times 10^5\ Па$

$T_{11} := T_3(p_3) = 305.422\ K$ $\quad p_{11} := p_3 = 8.372 \times 10^4\ Па$

$T_{12} := T_{11} = 305.422\ K$ $\quad p_{12} := p_3 = 8.372 \times 10^4\ Па$

$$N_t := G_1 \cdot c_p \cdot (T_2(p_3) - T_3(p_3)) = 1.505 \times 10^4\ Вт$$

Расчёт режима 2:

Баланс АТ1:

$$Q_{AT1} = G_1 \cdot c_p \cdot (T_1 - T_2)$$

$$Q_{AT1} = G_{12} \cdot c_p (T_5 - T_{12})$$

$\Delta p_{AT1} := 0.05 \cdot p_{tor2} = 3.64 \times 10^3\ Па$

$p_2 := p_{tor2} - \Delta p_{AT1} = 6.916 \times 10^4$ Па

Тогда:

$T_1 := T_{tor2}$

Баланс АТ2:

$G_4 = (1 - \beta) \cdot G_1$

$G_{bo} := \frac{Q_{co2}}{c_p \cdot (T_8 - T_{10})} = 0.437 \quad \frac{кг}{с}$

$G_8 = G_9 = G_{10} = G_{bo}$

Для вентиляора примем:

$T_9 := T_8 \cdot \left[1 + \frac{\pi_{v1}^{\frac{(k-1)}{k}} - 1}{\eta_{v1}} \right] = 330.71$ K

$N_{v1} := G_{bo} \cdot c_p \cdot (T_9 - T_8) = 3.327 \times 10^3$ Вт

$Q_{AT2} := Q_{co2} + N_{v1} = 7.727 \times 10^3$ Вт

$a := \frac{k - 1}{k}$

Опишем функции

$A(p) := 1 - \eta_t \cdot \left[1 - \left(\frac{p}{p_2} \right)^a \right]$

$T_2(p) := \frac{(1 - \beta \cdot \varepsilon_1) T_1}{1 - \beta \cdot \varepsilon_1 \cdot A(p)}$

$T_3(p) := A(p) \cdot T_2(p)$

$T_4(p, G) := T_3(p) + \frac{Q_{AT2}}{(1 - \beta) \cdot G \cdot c_p}$

$T_5(p) := T_3(p) + \varepsilon_1 \cdot (T_1 - T_3(p))$

$T_6(p, G) := (1 - \beta) \cdot T_4(p, G) + \beta \cdot T_5(p)$

Решим систему для p3 и G1 относительно АТ2 и вала:

$p_3 := 84000\ \text{Па}$ $G_1 := 0.41\ \frac{\text{кг}}{\text{с}}$ - первое прибижение

Given

$0 < p_3 < p_2$ $G_1 > 0$

$$Q_{AT2} = \varepsilon_2 \cdot c_p \cdot \min\left[G_{bo}, (1-\beta)\cdot G_1\right]\cdot\left(T_9 - T_3(p_3)\right)$$

$$\eta_м \cdot\left(T_2(p_3) - T_3(p_3)\right) = \frac{T_6(p_3, G_1)}{\eta_к}\cdot\left[\left(\frac{p_{occ2}}{0.9 p_3}\right)^a - 1\right]$$

$$\text{Find}(p_3, G_1) = \begin{pmatrix} 1.494\times 10^4 \\ 0.151 \end{pmatrix} \quad p_3 := 1.494\times 10^4 \quad G_1 := 0.151 \quad \Delta p_{AT2} := 0.1 p_3$$

Найдём температуры и давления в точках:

$T_1 = 366.588\ \text{K}$ $p_1 := p_{tor2} = 7.28\times 10^4\ \text{Па}$

$T_2(p_3) = 342.045\ \text{K}$ $p_2 = 6.916\times 10^4\ \text{Па}$

$T_3(p_3) = 251.089\ \text{K}$ $p_3 = 1.494\times 10^4\ \text{Па}$

$T_4(p_3, G_1) = 318.908\ \text{K}$ $p_4 := p_3 - \Delta p_{AT2} = 1.345\times 10^4\ \text{Па}$

$T_5(p_3) = 349.263\ \text{K}$ $p_5 := p_3 - \Delta p_{AT2} = 1.345\times 10^4\ \text{Па}$

$T_6(p_3, G_1) = 326.497\ \text{K}$ $p_6 := p_3 - \Delta p_{AT2} = 1.345\times 10^4\ \text{Па}$

$T_7 := T_6(p_3, G_1) + \left(\frac{T_6(p_3, G_1)}{\eta_к}\right)\cdot\left[\left(\frac{p_{occ2}}{0.9 p_3}\right)^a - 1\right] = 415.68\ \text{K}$ $p_7 := p_{occ2} = 2.582\times 10^4\ \text{Па}$

$T_8 = 323.15\ \text{K}$ $p_8 := p_{H0} = 1.013\times 10^5\ \text{Па}$

$T_9 = 330.71\ \text{K}$ $p_9 := 1.05\cdot p_8 = 1.064\times 10^5\ \text{Па}$

$T_{10} = 313.15\ \text{K}$ $p_{10} := p_{H0} = 1.013\times 10^5\ \text{Па}$

$T_{11} := T_3(p_3) = 251.089\ \text{K}$ $p_{11} := p_3 = 1.494\times 10^4\ \text{Па}$

$T_{12} := T_{11} = 251.089\ \text{K}$ $p_{12} := p_3 = 1.494\times 10^4\ \text{Па}$

$N_t := G_1\cdot c_p\cdot\left(T_2(p_3) - T_3(p_3)\right) = 1.382\times 10^4\ \text{Вт}$

Расчёт режима 3:

Баланс АТ1:

$$Q_{AT1} = G_1 \cdot c_p \cdot (T_1 - T_2)$$

$$Q_{AT1} = G_{12} \cdot c_p (T_5 - T_{12})$$

$$\Delta p_{AT1} := 0.05 \cdot p_{tor3} = 7.428 \times 10^3 \text{ Па}$$

$$p_2 := p_{tor3} - \Delta p_{AT1} = 1.411 \times 10^5 \text{ Па}$$

Тогда:

$$T_1 := T_{tor3}$$

Баланс АТ2:

$$G_4 = (1 - \beta) \cdot G_1$$

$$G_{bo} := \frac{Q_{co3}}{c_p \cdot (T_8 - T_{10})} = 0.477 \quad \frac{\text{кг}}{\text{с}}$$

$$G_8 = G_9 = G_{10} = G_{bo}$$

Для вентиляора примем:

$$T_9 := T_8 \cdot \left[1 + \frac{\pi_{v1}^{\frac{(k-1)}{k}} - 1}{\eta_{v1}} \right] = 330.71 \text{ K} \qquad N_{v1} := G_{bo} \cdot c_p \cdot (T_9 - T_8) = 3.629 \times 10^3 \text{ Вт}$$

$$Q_{AT2} := Q_{co3} + N_{v1} = 8.429 \times 10^3 \text{ Вт} \qquad a := \frac{k - 1}{k}$$

$$A(p) := 1 - \eta_t \cdot \left[1 - \left(\frac{p}{p_2} \right)^a \right]$$

Опишем функции

$$T_2(p) := \frac{(1 - \beta \cdot \varepsilon_1) T_1}{1 - \beta \cdot \varepsilon_1 \cdot A(p)}$$

$$T_3(p) := A(p) \cdot T_2(p)$$

$$T_4(p, G) := T_3(p) + \frac{Q_{AT2}}{(1 - \beta) \cdot G \cdot c_p}$$

$$T_5(p) := T_3(p) + \varepsilon_1 \cdot (T_1 - T_3(p))$$

$$T_6(p, G) := (1 - \beta) \cdot T_4(p, G) + \beta \cdot T_5(p)$$

Решим систему для p3 и G1 относительно АТ2 и вала:

$p_3 := 84000$ Па $G_1 := 0.41 \frac{кг}{с}$ - первое прибижение

Given

$0 < p_3 < p_2$ $G_1 > 0$

$$Q_{AT2} = \varepsilon_2 \cdot c_p \cdot \min\left[G_{bo}, (1 - \beta) \cdot G_1 \right] \cdot \left(T_9 - T_3(p_3) \right)$$

$$\eta_м \cdot \left(T_2(p_3) - T_3(p_3) \right) = \frac{T_6(p_3, G_1)}{\eta_к} \cdot \left[\left(\frac{p_{occ3}}{0.9 p_3} \right)^a - 1 \right]$$

$$\text{Find}(p_3, G_1) = \begin{pmatrix} 2.158 \times 10^4 \\ 0.225 \end{pmatrix}$$

$p_3 := 2.158 \times 10^4$ $G_1 := 0.225$ $\Delta p_{AT2} := 0.1 p_3$

Найдём температуры и давления в точках:

$T_1 = 428.493$ K $p_1 := p_{tor3} = 1.486 \times 10^5$ Па

$T_2(p_3) = 395.276$ K $p_2 = 1.411 \times 10^5$ Па

$T_3(p_3) = 272.177$ K $p_3 = 2.158 \times 10^4$ Па

$T_4(p_3, G_1) = 321.829$ K $p_4 := p_3 - \Delta p_{AT2} = 1.942 \times 10^4$ Па

$T_5(p_3) = 405.046$ K

$p_5 := p_3 - \Delta p_{AT2} = 1.942 \times 10^4$ Па

$T_6(p_3, G_1) = 342.633$ K

$p_6 := p_3 - \Delta p_{AT2} = 1.942 \times 10^4$ Па

$T_7 := T_6(p_3, G_1) + \left(\frac{T_6(p_3, G_1)}{\eta_к}\right) \cdot \left[\left(\frac{p_{occ3}}{0.9 p_3}\right)^a - 1\right] = 463.252$ K $\quad p_7 := p_{occ3} = 4.41 \times 10^4$ Па

$T_8 = 323.15$ K

$p_8 := p_{H0} = 1.013 \times 10^5$ Па

$T_9 = 330.71$ K

$p_9 := 1.05 \cdot p_8 = 1.064 \times 10^5$ Па

$T_{10} = 313.15$ K

$p_{10} := p_{H0} = 1.013 \times 10^5$ Па

$T_{11} := T_3(p_3) = 272.177$ K

$p_{11} := p_3 = 2.158 \times 10^4$ Па

$T_{12} := T_{11} = 272.177$ K

$p_{12} := p_3 = 2.158 \times 10^4$ Па

$N_т := G_1 \cdot c_p \cdot (T_2(p_3) - T_3(p_3)) = 2.786 \times 10^4$ Вт